

MULTIDETECTOR COMPUTED TOMOGRAPHIC AND LOW-FIELD MAGNETIC RESONANCE IMAGING ANATOMY OF THE QUADRIGEMINAL CISTERN AND CHARACTERIZATION OF SUPRACOLLICULAR FLUID ACCUMULATIONS IN DOGS

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Focal fluid accumulations in the supracollicular region are commonly termed quadrigeminal cysts and may be either subclinical or associated with neurologic deficits in dogs. Little published information is available on normal imaging anatomy and anatomic relationships for the canine quadrigeminal cistern. Objectives of this observational, cross-sectional study were to describe normal quadrigeminal cistern anatomy and determine the prevalence and characteristics of supracollicular fluid accumulations in dogs. Normal descriptions were accomplished using computed tomographic (CT) cisternography in one canine cadaver, and CT and magnetic resonance imaging (MRI) studies of the brain in four prospectively recruited dogs with no evidence of intracranial disease. Prevalence and characteristics descriptions were accomplished using a retrospective review of brain CT or MRI studies performed during the period of 2005–2015. The normal quadrigeminal cistern consistently exhibited a complex H shape and was separated from the third ventricle by a thin membrane. Prevalence of supracollicular fluid accumulations (SFAs) was 2.19% among CT studies ($n = 4427$) and 2.2% among MRI studies ($n = 626$). Dogs with SFA were significantly younger than control dogs ($P < 0.0001$). Shih-tzu (OR = 111.6), Chihuahua (OR = 81.1), and Maltese (OR = 27.6) breed dogs were predisposed ($P < 0.0001$). Among dogs with SFAs, the following three patterns were defined: (1) third ventricle (49.54%), (2) quadrigeminal cistern (13.51%), and (3) both third ventricle and quadrigeminal cistern (36.93%). Authors recommend that the term supracollicular fluid accumulation (SFA) should be used rather than the term quadrigeminal cyst to describe these focal fluid accumulations in dogs. © 2016 American College of Veterinary Radiology.

Key words: cisternography, computed tomography, magnetic resonance imaging, supracollicular fluid accumulation, quadrigeminal cistern.

Introduction

IN THE HUMAN ANATOMICAL LITERATURE, the term “quadrigeminal cistern” (also named *cisterna quadrigeminalis* or *cisterna venae magnae cerebri*) refers to the subarachnoid space located between the splenium of the corpus callosum and the upper portion of the cerebellum.^{1–4} The quadrigeminal cistern of the dog is mentioned in the major anatomy books,^{5,6} but is not reported in the *Nomina Anatomica Veterinaria* (NAV)⁷ and there are no published anatomical studies describing this structure to date. Despite the absence of any anatomical reference, the term *quadrigeminal cyst* is widely used in veterinary literature. There appears to be a

great confusion among various reports, with the same term used interchangeably to describe different intracranial cystic-like dilations in the quadrigeminal region of the dog and cat.^{8–14} However, the quadrigeminal cistern would not be always involved in these formations and the cystic nature not proven in most cases.^{8,11–14} Most recent works conducted using magnetic resonance imaging (MRI) in dogs documented different phenotypes of so-called quadrigeminal cysts, having different clinical impact: (1) quadrigeminal cistern arachnoid cyst, (2) caudodorsal diverticulum of the third ventricle, and (3) quadrigeminal cysts associated to the fourth ventricle.^{12,13,15} Accordingly, the term supracollicular fluid accumulations, rather than quadrigeminal cysts, especially in absence of histopathology confirmation, is more appropriate to describe these cystic-like lesions in imaging reports.^{12,13} Supracollicular fluid accumulations (SFAs) have been reported to have a prevalence of 0.7% in dogs, with male, brachycephalic, small breed dogs overrepresented.¹² Supracollicular fluid

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accumulations are often asymptomatic and may be found as incidental findings during brain imaging studies for other reasons.¹³ However, they may sometimes cause compression of the surrounding structures leading to severe neurologic manifestations. In most severe cases, a decompressive surgery may be indicated.⁹

A description of the normal imaging appearance of the quadrigeminal cistern in the dog is still lacking in veterinary radiology literature and imaging landmarks to distinguish various patterns of SFAs on routine computed tomography (CT) and MRI examinations have not been determined to date. We proposed here a hypothesis that some repetitive anatomic landmarks seen on imaging would allow to distinguish different patterns of supracollicular fluid accumulations.

Objectives of this study were (1) to assess the imaging anatomy of the quadrigeminal cistern in normal dogs; (2) to compare the results from normal subjects with imaging features in a sample of dogs with supracollicular fluid accumulations; (3) to determine the anatomical structures involved in various phenotypes; (3) to evaluate the prevalence of these conditions; and (4) to assess concomitant intracranial abnormalities in these dogs.

Material and Methods

The study design was observational and descriptive. The study consisted of two main steps, which will be described in this section. In order to determine the normal imaging anatomy appearance of the quadrigeminal region, in the first step of the study, we performed a cisterno-ventriculography in a cadaver and used the results as reference standard to interpret images of CT/MR examinations in normal dogs. The second step of the study was to assess the supracollicular fluid accumulation prevalence, epidemiology, and imaging characteristics in a large population of dogs underwent computed tomography or magnetic resonance imaging for various reasons.

Description of Normal Imaging Anatomy of the Quadrigeminal Region

To assess the normal imaging characteristics of the quadrigeminal region an MDCT cisterno-ventriculography was performed (GB) in a mesaticephalic dog from San Marco Veterinary Clinic. Upon owner request, the dog was euthanatized still on the CT table because of a poor prognosis. The dog did not present neurological signs or anatomical brain abnormalities on CT images. With owner consent, this dog was recruited for the MDCT cisterno-ventriculography, immediately after the euthanasia. Then, two mesaticephalic and two brachicephalic dogs underwent advanced brain imaging for various reasons having nor-

mal brain MDCT and MRI features were recruited to assess the normal quadrigeminal cistern appearance in living dogs. The quadrigeminal region was studied in these dogs using the MDCT cisterno-ventriculography as anatomical model.

Characterization of the Supracollicular Fluid Accumulations

Selection of MDCT cases. A retrospective search of the computer databases of the San Marco Veterinary Clinic (between 2005 and 2015) was conducted for dogs diagnosed with supracollicular fluid accumulations. Imaging database runs on a program (version 72, POA System 9.0-Copyright® 1993–2015) that allows users to access data by entering keywords related to the animal species, type of scanning, anatomy, imaging features, and final diagnosis. Keywords are entered into the system daily by radiologists. For this study, the keywords used were “quadrigeminal cyst,” “quadrigeminal cistern,” “intracranial arachnoid cyst,” “supracollicular fluid accumulations.” The inclusion criteria were that the patient had undergone a complete unenhanced and enhanced MDCT brain examination. Dogs with intracranial lesions in the mid and posterior fossa or history of head trauma were excluded from the study. Affected cases were included in the study group. For patients with repeat MDCT imaging, only the first examination was included in the study. The follow-up studies were only assessed for potential complication related to the presence of the supracollicular fluid accumulation. Concurrent intracranial/cranial anomalies, such as hydrocephalus, occipital dysplasia, or other were each time recorded. For all patients, history, clinical findings, and laboratory data were available.

Selection of MR cases. MR examinations of the brain performed at the Pingry Veterinary Hospital between 2010 and 2015 were reviewed. Similarly to that reported above for MDCT cases search, the MR reports run on the same program that allows data to be rapidly accessed by entering a keyword. Keyword used for this search was also similar to that used for MDCT cases search. Inclusion criteria were a whole brain MR examination including unenhanced and enhanced sequences. Further analyses with control group were not possible for MR population due to inconsistency in patients' data collected in the program.

MDCT and MR image analyses. Images obtained from MDCT and MRI examinations were retrieved from PACS (picture archiving and communication system) of both institutions, restored in a freestanding workstation (ADW 4.6 GE Medical Systems, Milano, Italy) or image processing software (OsiriX DICOM-viewer; Pixmeo,

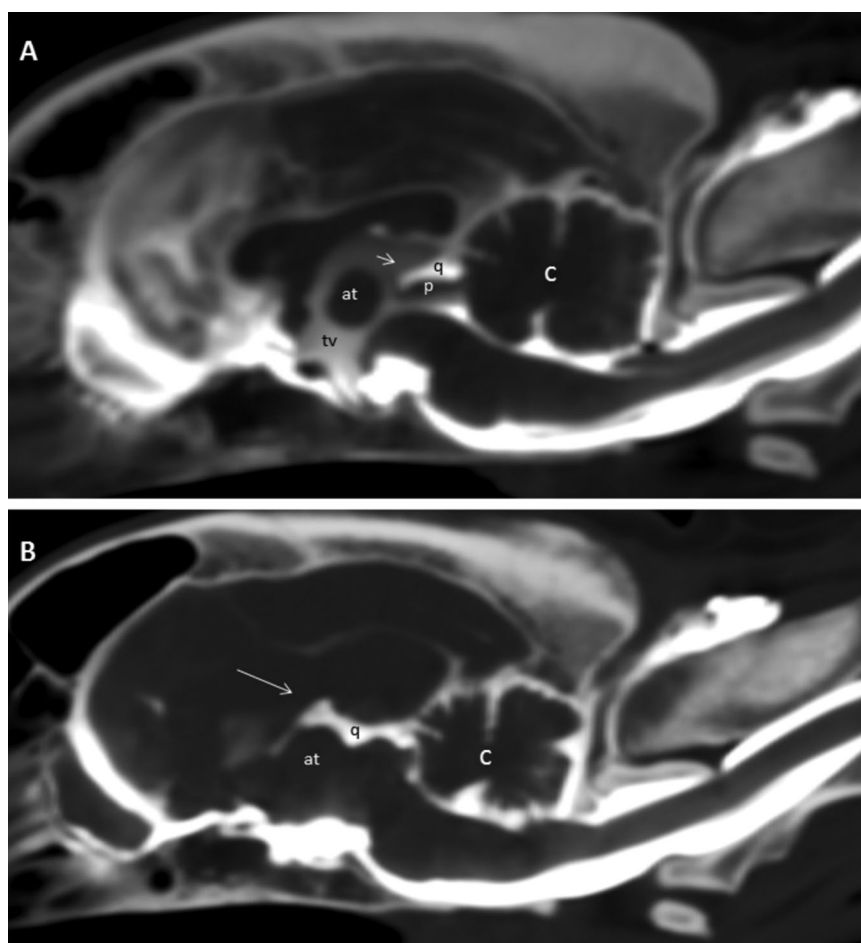


FIG. 1. Midsagittal (A) and right parasagittal (B) multiplanar reformatted MDCT images of the brain cisternoventriculography. The quadrige-minal cistern (q) is visible on midsagittal image as a short linear cavity on the roof of midbrain. Differences in postcontrast attenuation between the cisternal and ventricular systems highlights the anatomical relationship between the quadrige-minal cistern and the dorso-caudal part of the third ventricle (tv). A thin membrane isoattenuating to the soft tissue separate the quadrige-minal cistern from the overlying third ventricle (short arrow). On parasagittal image, the rostral extensions of the quadrige-minal cistern reach the middle-rostral part of the interthalamic adhesion (at) (long arrow); p, quadrige-minal plate.

Geneva, Switzerland). Postprocessing techniques, multiplanar reformatted (MPRs) and three-dimensional volume rendered (VR) images of the ventricular system, cisterns, and periencephalic subarachnoid space were used as an aid in analyzing the quadrige-minal cistern and its relationships with the surrounding structures. Images were reexamined for SFAs and a consensus was made between two authors (G.B. and M.R.) regarding CT and MR images interpretation. Based on previous description in veterinary literature, supracollicular fluid accumulations were defined “all those cystic-like structures dorsal to the quadrige-minal plate and rostral to the cerebellum” that contained a fluid with attenuation value on CT and signal intensity on MRI images comparable to that of the cerebrospinal fluid (CSF).^{10–15}

Various intracranial structures forming the supracollicular region were assessed on the basis of major anatomy books.^{5,6} The quadrige-minal cistern boundaries interpre-

tation was based on human imaging anatomy works and previous reports in veterinary literature.^{2,16}

Statistical analyses. Statistical analyses were performed using commercially available software (MedCalc for Windows version 12.4.0.0, MedCalc Software, Copyright ©1999–2013) by one author (M.C.). Normality of data was assessed with the Kolmogorov–Smirnov test. The Kolmogorov–Smirnov test showed that age ($P < 0.0001$) and body weight ($P < 0.0001$) were skewed. Univariate non-parametric variables were compared among groups of dogs via the Mann–Whitney test (age and bodyweight). All data were expressed as medians with ranges. The difference in sex frequency was analyzed using the Yates corrected χ^2 test. Breeds with 50 or more accessions and diagnosis of at least five dogs of that breed were compared with Mongrel dogs using the Yates corrected χ^2 test, and their association with outcome was expressed as odds ratios (ORs) with a 95%

confidence interval (CI). For gender, age, and body weight, the significance level was set at $P < 0.05$. Since more than one breed was tested, the P value for breeds was set at 0.01.

Finally, a multivariate logistic regression model was implemented with age, sex, and breed as predictors. Breed was included in the model grouping together Mongrel dogs and all the dogs belonging to other breed levels other than the significant ones at the univariate analysis, and using this group as reference level. Bodyweight was not included in the final model to avoid autocorrelation effect with breed.

Results

Description of Normal Imaging Anatomy of the Quadrigeminal Region

MDCT cisterno-ventriculography protocol and features. A solution of 4 ml of iodixanol (Visipaque 320 mg I/ml) and 4 ml of saline solution mixed in the same syringe was manually gently injected in the subarachnoid space via cisterna magna using a 25 gauge dedicated spinal needle for cerebrospinal fluid collection. Immediately after the contrast material injection, a brain CT scan was performed using a 16-MDCT scanner (Lightspeed 16, GE Medical Systems, Milan, Italy) with patient in sternal recumbency on the CT table with the CT protocol described below. Images of the cisterno-ventriculography showed markedly hyperattenuation of periencephalic subarachnoid spaces, which were site of primary injection. All cisternal spaces and cortical sulci were clearly visible as hyperattenuating layer all around the brain. Ventricular spaces showed diffuse hyperattenuation, but lower than that of the subarachnoid spaces, so that they could be easily distinguished from the cisternal system especially when they were in close connection each other. On midsagittal MPR images, the quadrigeminal cistern appeared as a small linear subarachnoid space just dorsal to the quadrigeminal plate (mesencephalic roof) (Fig. 1A). A thin membrane isoattenuating to brain parenchyma separated the quadrigeminal cistern from the third ventricle. On dorsal MPRs the quadrigeminal cistern appears as a large H-shaped layer whose extent was defined aborally by the pericerebellar subarachnoid space, laterally by the subarachnoid space that lined the medial part of the cerebral hemisphere and cranio-laterally by the subarachnoid space that lined the medial and aboral part of the two halves of the thalamus. Rostrally the quadrigeminal cistern gives rise to two extensions that border the dorso-aboral portion of the third ventricle on both sides. On parasagittal MPRs, these two extensions reached the dorso-lateral margin of the interthalamic adhesion (Fig. 1B and Fig. 2). On volume rendered images, the quadrigeminal cistern appeared of pyramidal shape with aboral base and cranial apex (Fig. 3A, B). Different imprints of the surrounding structures were evident on its surfaces: the aboral part

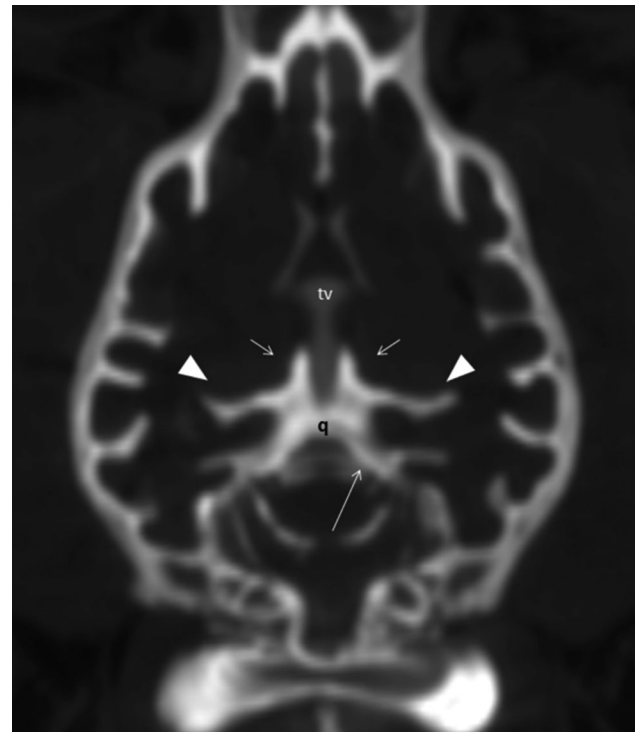


FIG. 2. Dorsal multiplanar reformatted MDCT image of the brain cisternoventriculography (dog 1). The quadrigeminal cistern (q) appears as a large H-shaped layer connected aborally with the pericerebellar subarachnoid space (long arrow) and laterally with the subarachnoid space, which lines the medial part of the cerebral hemispheres (arrowheads). Rostrally the quadrigeminal cistern gives rise to two extensions that border the dorso-aboral portion of the third ventricle (tv) on both sides (short arrows).

showed the imprint of the rostral part of the cerebellar vermis; the ventral surface showed imprints of rostral and caudal mesencephalic colliculi; the lateral walls showed thalamic imprints; the cranio-dorsal surface showed a linear concavity in which fits and lies the dorso-aboral portion of the third ventricle (third ventricle imprint) (Fig. 3B).

MDCT and MRI characteristics of quadrigeminal region of normal dogs. On MRI images of normal mesencephalic and brachicephalic dogs, the shape, and topography of the quadrigeminal cistern were comparable to the images of cisternography (Figs. 4 and 5). On unenhanced-MDCT images, the quadrigeminal cistern was clearly visible but poorly detailed. Dorsal MPR showed the same H-shape pattern as seen in the cisternography even if its rostral extensions and their relationships with the third ventricle were barely appreciable. On contrast-enhanced MDCT images the internal cerebral veins were evident on both side between the rostral extension of the quadrigeminal cistern (lining their medial margins) and the dorso-aboral part of the third ventricle. The aboral apex of the third ends at the convergence of the two internal cerebral veins in the great cerebral vein (Figs. 5 and 6). As seen on

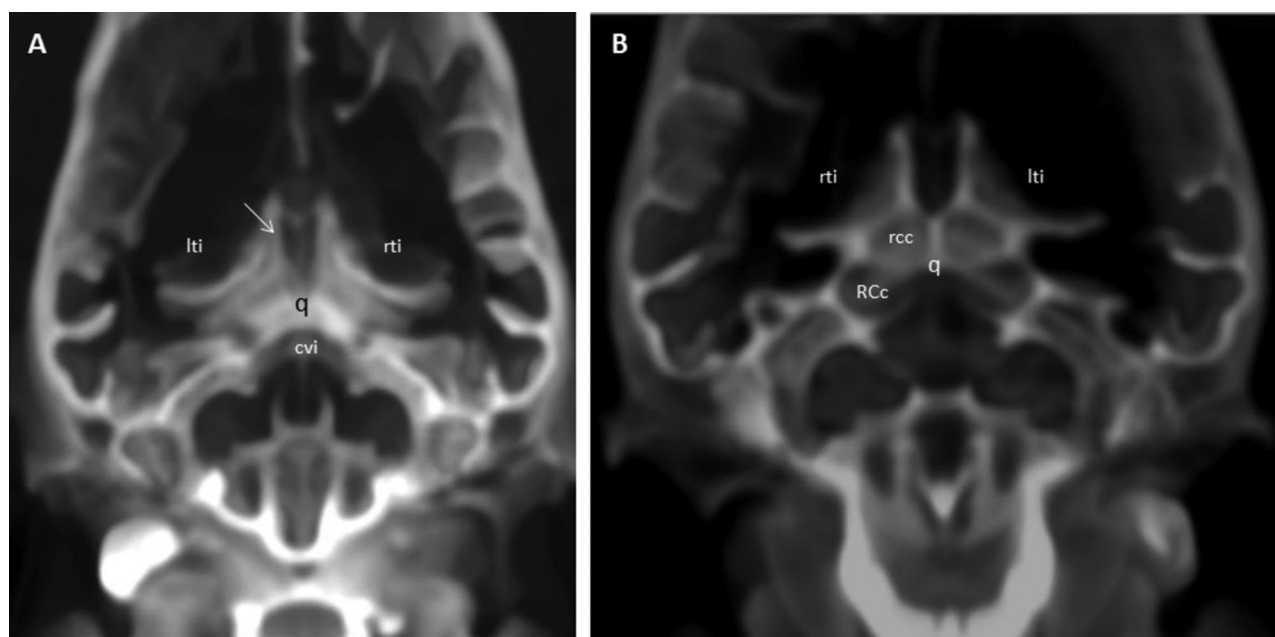


FIG. 3. Dorsal and ventral views (A, B) of 3D-volume rendered data obtained from CT-cisterno-ventriculography. The quadrigeminal cistern (q) showed pyramidal shaped with an aboral base and cranial apex. The aboral part shows the imprint of the rostral part of the cerebellar vermis (cvi); the ventral surface showed imprints of rostral and caudal mesencephalic colliculi (rcc: right cranial colliculus imprint; RCc: right caudal colliculus imprint); the lateral walls showed thalamic imprints (lti: left thalamic imprint; rti: right thalamic imprint); the cranio-dorsal surface shows a linear concavity in which fits and lies the dorso-aboral portion of the third ventricle (third ventricle imprint) (arrow).

cisternoventriculography, both in MRI and MDCT images the third ventricle and quadrigeminal cistern appeared separated by a thin membrane isointense and isoattenuating to brain parenchyma.

Characterization of the Supracollicular Fluid Accumulations

Description of the MDCT study population. One-hundred and nine reports with the select keywords were found from a total of 4427 contrast-enhanced brain scans of dogs underwent MDCT examination at the San Marco Veterinary Clinic in the specified period of time. Studies from 12 dogs were excluded because one of the following reasons: wrong diagnosis (3) and masses in the caudal fossa (2), hence included in the control group; repeat examinations of the same dog (7), definitively excluded from the study. A final total of 97 dogs (2.19% prevalence, 95% CI, 1.79–2.68) were included in the SFA-dogs population for further analyses. The remaining 4323 dogs without SFA represented the control group. Of the 4427 dogs, representing 145 breeds, 2276/4427 (51.4%) were males and 2151/4427 (48.6%) were females. Forty-eight out of the 2276 males (2.1%) and 49 out 2151 females (2.27%) had SFA and this difference was not statistically different ($P = 0.70$). The median age of dogs with SFA was 71 months (range 5–169 months), lower than the median age of dogs without SFA was 104 months (range 1–227 months). This difference was statistically sig-

nificant ($P = 0.0001$). The median bodyweight of dogs with SFA was 4.2 kg (range 0.9–26.5 kg), which was significantly lower than that of dogs without SFA (range 0.3–83 kg; $P = 0.0001$). Nineteen breeds were represented in the group of dogs with SFA. Three breeds showed an increased risk of having SFA: Shih-Tzu (OR = 111.6; CI, 95%; 44.2–281.7), Chihuahua (OR = 81.1; CI, 95%; 31.3–210.2), and Maltese (OR = 27.6; CI, 95%; 10.3–74.2).

Also the multivariate logistic regression model confirmed the significant effect of age (Wald = 9.06; $P = 0.03$; Exp(B) = 0.992; CI, 95%; 0.988–0.997) and breed (Wald = 253.03; $P < 0.0001$). The single breed levels showed an increased risk of having SFA as follows: Shih-Tzu (Exp(B) = 59.2; CI, 95%; 33.4–105.1), Chihuahua (Exp(B) = 30.5; CI, 95%; 16.2–57.5), and Maltese (Exp(B) = 12.9; CI, 95%; 6.6–25.2).

Description of the MR study population. Fourteen of 626 dogs underwent brain MR examination in the selected period of time met the inclusion criteria and were recruited for the study. The prevalence of dogs with SFA was 2.2% (95% CI, 1.28–3.82). The median body weight of MR-population was 7.5 kg (range 2–11 kg). Seven dogs were males and seven were females. The median age of dogs with SFA was 43 months (range 6–156 months). The represented breeds were: Chihuahua (5/14), Shih-Tzu (3/14), Mongrel (2/14), Maltese (1/14), Jack Russel (1/14), Pug (1/14), Cavalier King Charles Spaniel (1/14).

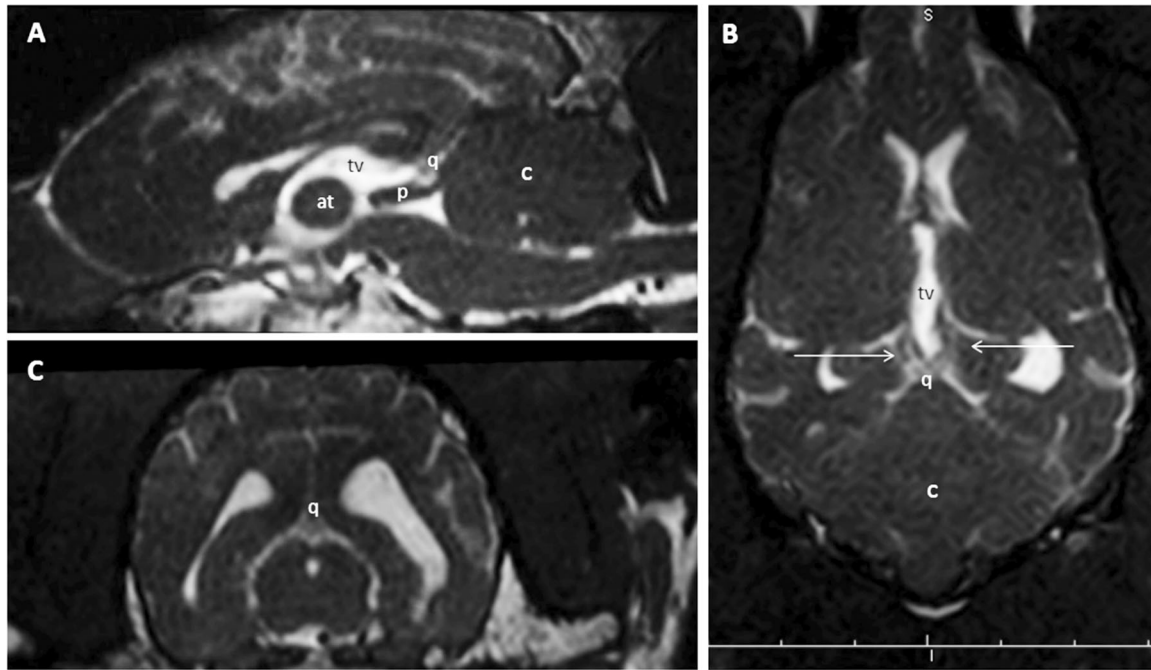


FIG. 4. Midsagittal (A), dorsal (B) and transverse (C) multiplanar reformatted MRI images (from a native dorsal balance steady-state free precession sequence) of the brain of a normal mesencephalic dog. Dorsal and transverse images were obtained at the level of the quadrigeminal cistern (q). On midsagittal MPR (A) image the quadrigeminal cistern appears as a small round subarachnoid space just dorsal to the quadrigeminal plate (p). Dorsal MPR (B) shows the same H-shape pattern as seen in the cisternography. The rostral extensions (long arrows) and their relationships with the third ventricle (tv) are clearly appreciable as in the MDCT images; at, interthalamic adhesion; c, cerebellum.

Protocol for MDCT scans. MDCT examinations were performed with 16-MDCT scanner (Lightspeed 16, GE Medical Systems, Milan, Italy). MDCT scanning and reconstruction parameters were as follows: axial modality, 2 s/gantry rotation, detector configuration 16×0.625 , 10 mm increment, 0.625 mm slice thickness; dose parameters 120 kV, 300 mAs. The contrast medium protocol was: 2 ml/kg ml of iodixanol 320 mgI/ml injected intravenously at a 3 ml/s followed by a saline flush, using a dual barrel injector CT system (Stellant, Medrad, Milano, Italy).

Protocol for MRI scans. MRI of the brain was performed with the dog under general anesthesia using a 0.25 Tesla permanent magnet (ESAOTE VET-MR GRANDE, Esaote, Genova, Italy). MRI sequences protocol included Fast Spin Echo T2-weighted images acquired in sagittal and transverse plane, a FLAIR acquired in the transverse plane, and Spin Echo T1-weighted images acquired in the transverse plane before and after intravenous administration of paramagnetic contrast medium (gadopentate dimeglumine 500 mmol/ml; dosage 0.15 mmol/kg). A balance steady-state free precession sequence (BSSFPs) was added to the standard sequence protocol, before contrast medium administration, in order to obtain a detailed volumetric dataset of the brain with bright CSF signal.

MDCT and MRI characteristics of the supracollicular fluid accumulations and concomitant findings. In all 111 examinations (97 MDCT and 14 MRI), the thin membrane isointense or isointense to the brain parenchyma, respectively on MDCT or MRI images, was always visible. This membrane clearly separated the fluid accumulation in two compartments, the more rostral III ventricle and the more caudal quadrigeminal cistern. The size of the two compartments varied depending on the position of the thin membrane in respect to the quadrigeminal plate. Three repetitive patterns of supracollicular fluid accumulation were identified (Fig. 7) and classified based on the anatomy reference provided by the cisterno-ventriculography and normal subjects' analysis:

Pattern 1 (55/111 dogs, 49 from MDCT database and six from MRI database): SFA consisted of a dorsocaudal expansion of the third ventricle with normal or minimally enlarged quadrigeminal cistern (SFA-III).

Pattern 2 (41/111 dogs, 36 MDCT and five MRI) had a SFA consisting of an expansion of both the third ventricle and the quadrigeminal cistern (SFA III-QC).

Pattern 3 (15/111 dogs, 12 MDCT and three MRI) had a SFA consisting of an expansion of the sole quadrigeminal cistern with normal third ventricle appearance (SFA-QC).

Seventy-seven/111 dogs underwent MDCT or MR examination showed lateral ventricles enlargement.

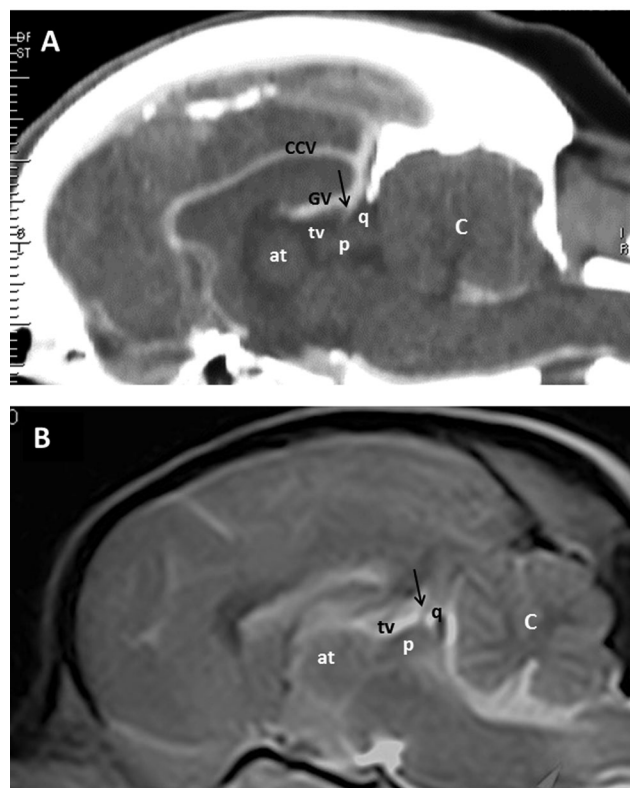


FIG. 5. Normal appearance of the supracollicular region in two Shih-Tzu dogs without SFA. (A) Midsagittal multiplanar reformatted MDCT image of the brain. (B) Midsagittal T2-weighted MRI image of the brain. The caudal part of the third ventricle (tv) normally extends caudally lying on the caudal half of the quadrigeminal plate (p), next to the quadrigeminal cistern (q). The separation between third ventricle and quadrigeminal cistern is evident (arrows). (C), cerebellum; CCV, vein of corpus callosum; GV, great cerebral vein; at, interthalamic adhesion.

Concomitant cranial anomalies were evaluated in 97/111 dogs underwent MDCT examination: 34/97 dogs showed underdevelopment of the occipital bone and/or tentorium cerebelli. Cerebellar compression due to SFA expansion was found in 12/111 dogs. Of these compressive SFAs 8/12 were attributable to Pattern 3 (SFA-QC), 3/12 to Pattern 2 (SFA III-QC), and one to Pattern 1 (SFA-III).

Discussion

Prevalence of supracollicular fluid accumulation in our study was 2.19% and 2.2% in the MDCT-population and MR population respectively, which is higher than the prevalence of SFA reported in a previous study (0.68%).¹² Breeds involved in the two studies are diverse and this might explain the higher prevalence we documented here. Dogs with SFA showed a lower body weight when compared with dogs without SFA ($P = 0.0001$). This is probably because small and toy breeds are more likely to have SFA. In our study, Shih-Tzu (OR = 111.6), Chihuahua (OR = 81.1), and Maltese (OR = 27.6) showed a breed predisposition to have SFA. Among these breeds, Shih-Tzu dogs were over-

represented in a previous study as well.¹² However, since the absence of a control group in that study, an increased risk of this breed to have SFA was not proven.

The evaluation of clinical signs related to the presence of SFAs was not an aim of the present study. Nevertheless, is interesting to note that in our study population the median age of dogs with SFA was significantly lower to that of dogs without SFA (71 months and 104 months, respectively) ($P = 0.0001$). It could be argued that SFAs may be responsible of some clinical effects that required advanced imaging examination. More likely, the same breeds also have predisposition for other congenital pathological conditions having some clinical impact. This point should be investigated in future studies.

Previous studies with MRI documented three different patterns of “quadrigeminal cysts” associated with third ventricle or fourth ventricle, or a loculated phenotype appearing to have relation to both ventricles.^{12,13} However, in the first step of our study, we demonstrated with CT-cisterno-ventriculography that there are not communications between the quadrigeminal cistern and the ventricular structures. A thin membrane between third ventricle and quadrigeminal cistern seen in normal dogs was always visible also in dogs with SFA providing a landmark of separation between the two compartments. This thin membrane, isoattenuating to soft tissues on MDCT and isointense on MRI to brain parenchyma, was best appreciable on midsagittal plane. It most likely represented the third ventricle wall, where the pia mater join the ependymal layer forming the tela choroidea of the third ventricle⁵ (Fig. 1A). The membrane has been reported to consist of a set of anatomical structures: cranial and caudal stems of the pineal stalk, habenular commissure, posterior commissure, and vascular structures (from the caudal cerebral artery and tributaries of the great cerebral vein).^{5,15}

This consistently detected landmark of separation was used to define three repeatable different patterns of SFAs that deviated from the previous classification.¹² Most of the supracollicular fluid accumulations seen in our patients were attributable to the dorsocaudal outpocketing of the third ventricle (SFA-III), and this was consistent with previous studies. A dorsocaudal diverticulum of the third ventricle has been also detected together with an enlargement of the quadrigeminal cistern in our and previous studies (SFA III-QC).^{12,13} None of our cases showed a communication between this two compartments. These findings supported the authors’ recommendation that the term supracollicular fluid accumulation should be used rather than quadrigeminal cyst to describe these imaging features, as previously suggested.¹² A few cases of the present study showed pseudo-cystic enlargement of the sole quadrigeminal cistern (SFA-QC) and none showed any communication with the third ventricle, aqueducts, or the fourth ventricle.

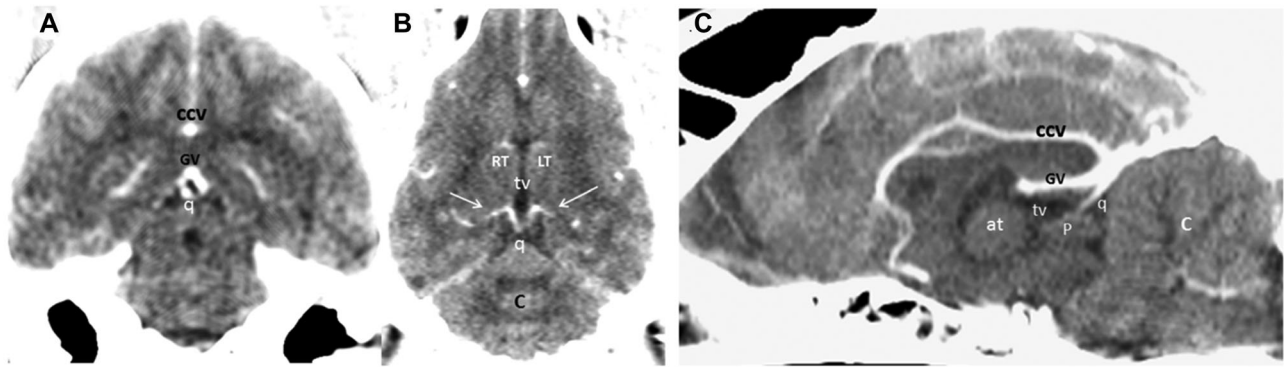


FIG. 6. Transverse (A), dorsal (B), and midsagittal (C) multiplanar reformatted contrast-enhanced MDCT images of the brain of a normal mesocephalic dog. Dorsal and transverse images were obtained at the level of the quadrigeminal cistern (q). On midsagittal MPR image the quadrigeminal cistern (arrowhead) appears as a small triangular subarachnoid space just dorsal to the quadrigeminal plate (C). Dorsal MPR showed the same H-shape pattern as seen in the cisternography. In the dorsal plane (B) the internal cerebral veins (white arrows) are evident on both side between the rostral extension of the quadrigeminal cistern (lining their medial margins) and the dorso-aboral part of the third ventricle (tv). The aboral apex of the third ventricle ends at the convergence of the two internal cerebral veins in the great cerebral vein (GV - not visible on dorsal MPR). p, quadrigeminal plate; at, Interthalamic adhesion; tv, third ventricle; RT, right thalamus; LT, left thalamus; CCV, vein of corpus callosum.

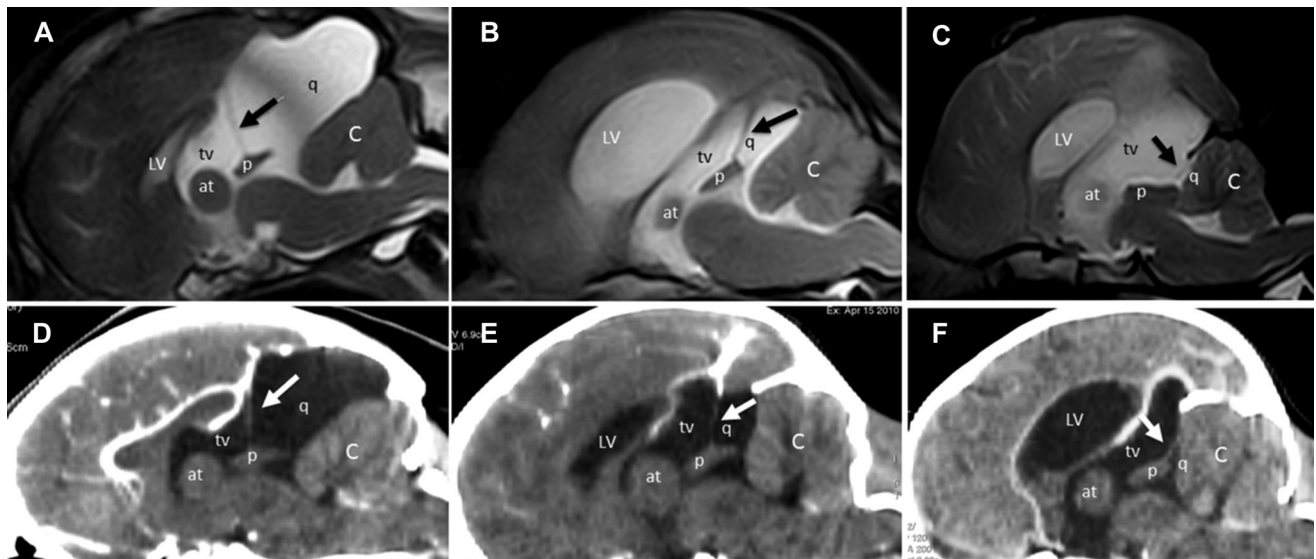


FIG. 7. Different patterns of SFAs as seen on midsagittal MR (A, B, C) and MDCT (D, E, F) images of the brain. (A) Balance steady-state free precession sequence of the brain of a 2-year-old male Pug (dog 4) with dilation of the quadrigeminal cistern (q). (B) T2-weighted image of the brain of a 4-year-old male Chihuahua (dog 5), with moderate dorsocaudal dilation of the third ventricle (tv) and moderate dilation of the quadrigeminal cistern. Lateral ventricles (LV) are also enlarged. (C) T2-weighted image of the brain of a 1-year-old male Mongrel with large dorsocaudal expansion of the III ventricle. Moderate enlargement of the lateral ventricles (LV). (D) MDCT image of the brain in a 11-year-old, female, Shih-Tzu with dilation of the quadrigeminal cistern. (E) Midsagittal view in a 2-year-old, male Chihuahua with dorsocaudal outpocketing of the III ventricle and moderate enlargement of the quadrigeminal cistern. (F) Dorsocaudal supracollicular dilation only of the III ventricle in a 8-months-old, male, Yorkshire Terrier. In each image, note the thin membrane that separates the III ventricle from the quadrigeminal cistern (arrows) and the displacement of the quadrigeminal plate (p). Enlargement of the quadrigeminal cistern leads to the dorsoventral compression of the cerebellum (C). Enlargement of III ventricle leads to an anteroposterior cerebellar compression. Large dilations of quadrigeminal cistern and third ventricle lead to displacement of the occipital lobes (A, C, D, F); at, interthalamic adhesion.

In a previous study in dogs, obstructive hydrocephalus secondary to cerebellar compression and outflow obstruction at the level of the IV ventricle and mesencephalic aqueduct was reported in three out of 28 dogs with SFAs included in that study.¹² In our study population, 77/111 dogs (70%) showed prominent lateral ventricles. In all these patients, the ventriculomegaly was interpreted to be breed-related rather than an obstructive hydrocephalus secondary

to the SFA. Patterns 1 (SFA-III) and 2 (SFA III-QC), were those mostly associated with various degree of lateral ventricles enlargement. Authors hypothesize that a dorso-caudal diverticulum of the third ventricle, in predisposed breeds, would most likely be part of total congenital hydrocephalus rather than an anomaly itself.

Compression of the cerebral occipital lobes and the cerebellum has been described in association with SFAs in

dogs.^{12,13,17,18} In our series, 12/111 dogs (10.9%) showed compression of the cerebellum as a consequence of SFA expansion. In dogs with Pattern 1 (SFA-III), the dorso-caudal expansion of the III ventricle may cause a slight compression of the cranial margin of the cerebellum (Fig. 7). Most of the cerebellar compression (8/12) was found in dogs with pattern 3 (SFA-QC) that showed a dorso-ventral compression of the cerebellum (Fig. 7). In a previous published study in 28 dogs, to evaluate the percentage of cerebral and cerebellar compression, diameters of the cerebellum were assessed on sagittal MR images using the osseous tentorium cerebelli as landmark.¹² However, 25/97 dogs underwent MDCT in the present study (22.7%) and these showed an underdeveloped osseous tentorium cerebelli with a vertical growth. Accordingly, the osseous tentorium cerebelli may not be the ideal landmark for cerebellar compression assessment. The osseous tentorium cerebelli hypoplasia has been reported in several cases of congenital cystic disorders of the posterior fossa in humans such as Blake's pouch cyst, Dandy-Walker malformation, and arachnoid cysts.^{19–21} The expansion of these fluid-filled spaces during the early fetal period leads to dehiscence of the developing tentorium cerebelli. Similarly, authors hypothesized that the reduced size of the osseous tentorium in our group of dogs would support the congenital origin of SFAs.

In our study, 23/97 dogs that underwent MDCT examination (20.9%) showed occipital dysplasia. This malformation is defined as an abnormally large foramen magnum caused by incomplete ossification of the ventromedial supraoccipital bone during gestation.²² Authors did not hypothesize any anatomical relationship between SFA and occipital dysplasia. Small and toy breeds are commonly reported to have occipital dysplasia. We considered this structural abnormality as a concomitant, breed-related congenital disorder.

An important limitation of the present study was that CT-cisterno-ventriculography was only performed in a

mesencephalic dog and not in subjects of predisposed breeds, such as Shih-Tzu that might have some anatomical differences. However, this study provided preliminary information of normal landmarks of the quadrigeminal cistern in dogs as it appears on routine clinical imaging and allowed the introduction of three different patterns of SFAs.

Anecdotally, authors considered MDCT cisterno-ventriculography to provide detailed anatomic information of the quadrigeminal cistern and its relationships with surrounding structures and this helped their interpretation on routine imaging. Authors considered MRI to provide excellent images of the quadrigeminal cistern comparable to those obtained with MDCT-cisterno ventriculography. High contrast resolution and thin-slice MDCT images of the brain were necessary to delineate these structures in normal dogs on various spatial planes. The balance steady-state free precession MR sequence was considered to be particularly useful in this study to obtain high quality MPR images of the subarachnoid spaces and overcome the limits of low spatial resolution of this imaging technique when compared to isotropic MDCT techniques. Authors recommend that this sequence be added to a standard sequence protocol for brain MRI in cases with supracollicular lesions. Thin-section MDCT provided further information regarding the cranial/intracranial concomitant anomalies. Therefore, authors propose that combining MDCT and MRI may be helpful for planning interventional approaches in symptomatic patients.

In conclusion, findings from the current study clarified anatomy and anatomic relationships for the canine quadrigeminal cistern. Three patterns of SFAs were identified and described: (1) third ventricle, (2) quadrigeminal cistern, and (3) both third ventricle and quadrigeminal cistern. Authors recommend that the term supracollicular fluid accumulation (SFA) should be used rather than the term quadrigeminal cyst to describe these focal fluid accumulations in dogs.

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